

BASIS FOR DECISION MEMO

Permit Processor: Bradley Popovich
 Date: August 2, 2018
 Permit No. MI0020257
 Facility's Designated Name: Cadillac WWTP

Monitoring Point 001A: Authorization to discharge 3.2 MGD of treated municipal wastewater from Monitoring Point 001A through Outfall 001. Outfall 001 discharges to the Clam River.

	<u>Maximum Limits for Quantity or Loading</u>				<u>Maximum Limits for Quality or Concentration</u>				<u>Monitoring Frequency</u>	<u>Sample Type</u>	<u>Basis for Limits</u>
<u>Parameter</u>	<u>Monthly</u>	<u>7-Day</u>	<u>Daily</u>	<u>Units</u>	<u>Monthly</u>	<u>7-Day</u>	<u>Daily</u>	<u>Units</u>			
Flow	(report)	---	(report)	MGD	---	---	---	---	Daily	Report Total Daily Flow	PWJ
Carbonaceous Biochemical Oxygen Demand (CBOD ₅)											
June – September <i>May – November</i>	190	270	(report)	lbs/day	7	---	10	mg/l	5x Weekly	24-Hr Composite	WQBEL/AD
October – November	290	480	(report)	lbs/day	11	---	17	mg/l	5x Weekly	24-Hr Composite	PWJ
December – March	670	910	(report)	lbs/day	25	---	34	mg/l	5x Weekly	24-Hr Composite	WQBEL/AD
April – May / April	450	670	(report)	lbs/day	17	---	25	mg/l	5x Weekly	24-Hr Composite	WQBEL/AD
Total Suspended Solids (TSS)											
June – September <i>May – November</i>	530	800	(report)	lbs/day	20	30	(report)	mg/l	5x Weekly	24-Hr Composite	PWJ
October – May <i>December – April</i>	800	1200	(report)	lbs/day	30	45	(report)	mg/l	5x Weekly	24-Hr Composite	PWJ
Ammonia Nitrogen (as N)											
June – September <i>May – November</i>	24	53	(report)	lbs/day	0.9	---	2.0	mg/l	5x Weekly	24-Hr Composite	WQBEL/AD
October – November	88	---	(report)	lbs/day	1.6	---	2.8	mg/l	5x Weekly	24-Hr Composite	PWJ
December – March	99	---	(report)	lbs/day	3.7	---	(report)	mg/l	5x Weekly	24-Hr Composite	WQBEL/AD
April – May / April	130	---	(report)	lbs/day	4.8	---	(report)	mg/l	5x Weekly	24-Hr Composite	WQBEL/AD
Total Phosphorus (as P)											
May – March	13	---	(report)	lbs/day	0.5	---	(report)	mg/l	5x Weekly	24-Hr Composite	WQBEL

<u>Parameter</u>	<u>Monthly</u>		<u>Daily</u>	<u>Units</u>	<u>Monthly</u>		<u>Daily</u>	<u>Units</u>	<u>Monitoring Frequency</u>	<u>Sample Type</u>	<u>Basis for Limits</u>
April	17	---	(report)	lbs/day	0.65	---	(report)	mg/l	5x Weekly	24-Hr Composite	WQBEL
Fecal Coliform Bacteria	---	---	---	---	200	400	(report)	cts/100 ml	5x Weekly	Grab	WQS
Perfluorooctane sulfonate (PFOS)	(report)	---	(report)	lbs/day	(report)	---	(report)	ng/l	2x Annually	Grab	WQC
Perfluorooctanoic acid (PFOA)	(report)	---	(report)	lbs/day	(report)	---	(report)	ng/l	2x Annually	Grab	WQC
Acute Toxicity	---	---	---	---	---	---	(report)	TU _A	Quarterly	24-Hr Composite	
Chronic Toxicity	---	---	---	---	1.1	---	---	TU _C	Quarterly	24-Hr Composite	
Available Cyanide	0.16	---	---	lbs/day	5.9	---	(report)	ug/l	Monthly	Grab	
Total Copper	0.3	---	0.82	lbs/day	11	---	31/(report)	ug/l	Weekly Quarterly	24-Hr Composite	WQBEL
Total Nickel	(report)	---	(report)	lbs/day	(report)	---	(report)	ug/l	Quarterly	24-Hr Composite	WQBEL
Total Selenium	0.15	---	(report)	lbs/day	5.7	---	(report)	ug/l	Monthly	24-Hr Composite	WQBEL
Bis(2-ethylhexyl)phthalate	0.82	---	(report)	lbs/day	31	---	(report)	ug/l	Monthly	24-Hr Composite	WQBEL
Total Mercury	(report)	---	---	lbs/day	(report)	---	---	ng/l	Quarterly	Grab	
Corrected	(report)	---	(report)	lbs/day	(report)	---	(report)	ng/l	Quarterly	Calculation	WQC
Uncorrected	---	---	---	---	---	---	(report)	ng/l	Quarterly	Grab	WQC
Field Duplicate	---	---	---	---	---	---	(report)	ng/l	Quarterly	Grab	WQC
Field Blank	---	---	---	---	---	---	(report)	ng/l	Quarterly	Preparation	WQC
Laboratory Method Blank	---	---	---	---	---	---	(report)	ng/l	Quarterly	Preparation	WQC
	<u>12-Month Rolling Avg</u>				<u>12-Month Rolling Avg</u>						
Total Mercury	0.000053	---	---	lbs/day	2.0	---	---	ng/l	Quarterly	Calculation	WQV
					<u>Minimum % Monthly</u>		<u>Minimum % Daily</u>				
TSS Minimum % Removal											
October – May December – April	---	---	---	---	85	---	(report)	%	Monthly	Calculation	STS

<u>Parameter</u>					<u>Minimum Daily</u>		<u>Maximum Daily</u>	<u>Units</u>	<u>Monitoring Frequency</u>	<u>Sample Type</u>	<u>Basis for Limits</u>
pH	---	---	---	---	6.5	---	9.0	S.U.	5x Weekly <i>Daily</i>	Grab	WQS
Dissolved Oxygen											
May – November	---	---	---	---	6.0	---	---	mg/l	5x Weekly	Grab	WQBEL/AD
December – April	---	---	---	---	5.0	---	---	mg/l	5x Weekly	Grab	WQBEL/AD

PERMIT CONDITIONS:

Final Effluent Limitations, Monitoring Point 001A

Quantification Levels and Analytical Methods for Selected Parameters

Additional Monitoring Requirements

Pollutant Minimization Program for Total Mercury

Pollutant Minimization and Source Evaluation Program for Perfluorooctane Sulfonate (PFOS) and/or Perfluorooctanoic Acid (PFOA)

Untreated or Partially Treated Sewage Discharge Reporting and Testing Requirements

Facility Contact

Monthly Operating Reports

Asset Management

Discharge Monitoring Report – Quality Assurance Study Program

Schedule of Compliance (not required)

Storm Water Pollution Prevention (not required)

Michigan Industrial Pretreatment Program

Residuals Management Program for Land Application of Biosolids

NOTES:

The WQBEL Toxic memo recommended monitoring for the following parameters: 3,3-Dichlorobenzidine, Benzidine, Hexachlorobenzene, Hexachlorobutadiene, Hexachlorocyclopentadiene, Acrylonitrile, 4-Chloro-3-methylphenol, Fluoranthene, Pentachlorophenol, Phenanthrene, and 2,4,6-Trichlorophenol. These parameters have not been included in the draft permit because they are not expected in the discharge, however, the compounds and appropriate quantification levels (QL's) are now listed in Part I.A.2, and these parameters are included in the additional monitoring requirements under Part I.A.3.

Less stringent limits are recommended for Dissolved Oxygen for Monitoring Point 001A. Based on a review of the facility's existing effluent quality, Permits Section determined that it is appropriate to retain the current limits based on antidegradation.

Less stringent limits are recommended for CBOD₅, and Ammonia Nitrogen (as N) for the months of December through September for Monitoring Point 001A. Based on a review of the facility's existing effluent quality, Permits Section determined that it is appropriate to retain the current limits based on antidegradation.

The facility is in the process of applying for a No Exposure Certification (NEC) that will cover any storm water discharges during this permit reissuance cycle.

The requirement to report total mercury results and associated data for corrected, uncorrected, field duplicate, field blank, and laboratory method blank results is added in order to be consistent with the sampling and QA/QC requirements of EPA Method 1631, Revision E, "*Mercury in Water by Oxidation, Purge and Trap, and Cold Vapor Atomic Fluorescence Spectrometry*".

Permits Section followed up with the facility and the facility confirmed that they are no longer using chlorine for disinfection.

An analysis of the facility's existing data was conducted, and it was determined that 99% of the time the facility is meeting the daily PWJ limits for CBOD₅ and Ammonia Nitrogen (as N) for the months of October through November. The completed analysis also found that 95% of the time the facility is meeting the monthly PWJ limit for Ammonia Nitrogen (as N) for the months of October through November).

The facility requested alternate seasons be evaluated within the WQBEL – conventional modeling due to weather factors within their northern geographical location. Alternate seasons were modeled and incorporated within the draft NPDES permit.

Limit Change Key

Normal Type = existing requirement - carried over from previous version of permit

Bold Type = new requirement - not in previous version of permit

Italic = deleted requirement - not carried over from previous version of permit

Basis for Limits Key

WQBEL - Water Quality-Based Effluent Limit

STS - Secondary Treatment Standard

WQC - Water Quality Concern

WQS - Water Quality Standard

WQV - Water Quality Variance

PWJ - Permit Writer's Judgment

AD – Antidegradation